

Module 7: Photosynthesis & Cellular Respiration — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Write and explain the overall equations for both photosynthesis and cellular respiration
2. Recognize that photosynthesis and respiration are complementary — reverse equations
3. Describe the structure of chloroplasts and mitochondria and where each stage occurs
4. Compare the light-dependent reactions and the Calvin Cycle in photosynthesis
5. Compare glycolysis, the Krebs Cycle, and the Electron Transport Chain in cellular respiration
6. Explain chemiosmosis and ATP synthase in both processes
7. Compare aerobic respiration with anaerobic fermentation
8. Trace the flow of energy from sunlight → glucose → ATP → cellular work

The Big Picture: Two Sides of One Equation

Photosynthesis and cellular respiration are mirror images:

	Photosynthesis	Cellular Respiration
Equation	$6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$	$\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{ATP (energy)}$
Type	Anabolic (builds glucose)	Catabolic (breaks down glucose)
Energy	Captures light energy $\rightarrow$ chemical energy	Releases chemical energy $\rightarrow$ ATP
Organelle	Chloroplast	Mitochondria
Organisms	Plants, algae, cyanobacteria	Nearly all living organisms
O₂	Produced (waste product)	Consumed (final electron acceptor)
CO₂	Consumed (raw material)	Produced (waste product)

The glucose produced by photosynthesis is the fuel consumed by cellular respiration. The O₂ released by photosynthesis is the O₂ used by respiration. The CO₂ released by respiration is the CO₂ used by photosynthesis. **These two processes cycle matter and energy through the biosphere.**

Key Terms to Know

Photosynthesis Terms

- **Chloroplast** — Organelle where photosynthesis occurs (double membrane)
- **Thylakoid** — Membrane-bound sacs inside chloroplast; site of light reactions
- **Granum** — Stack of thylakoids
- **Stroma** — Fluid surrounding thylakoids; site of Calvin Cycle
- **Chlorophyll** — Green pigment that absorbs red and blue light
- **Photosystem I and II** — Protein complexes in thylakoid that capture light energy
- **Light-Dependent Reactions** — Occur in thylakoid; split water, produce ATP + NADPH + O₂

- **Calvin Cycle (Light-Independent Reactions)** — Occur in stroma; use ATP + NADPH to fix CO₂ into G3P (sugar)
- **RuBisCO** — Enzyme that fixes CO₂ in the Calvin Cycle
- **C₃ Plants** — Standard Calvin Cycle pathway (most plants)
- **C₄ Plants** — Separate carbon fixation spatially to avoid photorespiration (corn, sugarcane)
- **CAM Plants** — Separate carbon fixation temporally — open stomata at night (cacti, succulents)

Cellular Respiration Terms

- **Mitochondria** — Organelle where cellular respiration occurs (double membrane)
- **Cristae** — Inner mitochondrial membrane folds; site of Electron Transport Chain
- **Matrix** — Inner fluid of mitochondria; site of Krebs Cycle
- **Glycolysis** — Splitting of glucose ($6C \rightarrow 2 \text{ pyruvate} + 2 \text{ ATP} + 2 \text{ NADH}$); occurs in cytoplasm
- **Pyruvate Oxidation** — Pyruvate enters mitochondria, converted to Acetyl-CoA + CO₂ + NADH
- **Krebs Cycle (Citric Acid Cycle)** — In matrix; oxidizes Acetyl-CoA, producing NADH, FADH₂, ATP, CO₂
- **Electron Transport Chain (ETC)** — In cristae; electrons from NADH/FADH₂ flow through protein complexes
- **Oxidative Phosphorylation** — ATP production driven by the proton gradient (chemiosmosis) at the ETC
- **ATP Synthase** — Enzyme that generates ATP as H⁺ ions flow down their gradient
- **Chemiosmosis** — The flow of H⁺ ions across a membrane through ATP synthase to make ATP
- **Fermentation** — Anaerobic pathway that regenerates NAD⁺ so glycolysis can continue
- **Lactic acid fermentation** — pyruvate $\rightarrow$ lactate (muscle cells, some bacteria)
- **Alcohol fermentation** — pyruvate $\rightarrow$ ethanol + CO₂ (yeast)

Photosynthesis: Stage by Stage

Stage 1: Light-Dependent Reactions (Thylakoid Membrane)

Input	Output
H ₂ O, light energy	ATP, NADPH, O ₂

- Light hits **Photosystem II** → water is split ($\text{H}_2\text{O} \rightarrow 2\text{H}^+ + \frac{1}{2}\text{O}_2 + 2\text{e}^-$)
- Electrons pass through the **electron transport chain** → energy pumps H^+ into thylakoid
- H^+ flows back out through **ATP synthase** → ATP is made (chemiosmosis)
- Electrons reach **Photosystem I** → re-energized by light → used to make **NADPH**

Stage 2: Calvin Cycle (Stroma)

Input	Output
CO ₂ , ATP, NADPH	G3P (3-carbon sugar), ADP, NADP ⁺

- **Carbon fixation:** RuBisCO attaches CO₂ to a 5-carbon molecule (RuBP)
- **Reduction:** ATP and NADPH convert the product to G3P
- **Regeneration:** Some G3P regenerates RuBP; the rest is used to build glucose
- 3 turns of the cycle fix 3 CO₂ and produce 1 net G3P; **6 turns = 1 glucose**

Cellular Respiration: Stage by Stage

Stage 1: Glycolysis (Cytoplasm)

Input	Output
1 Glucose (6C)	2 Pyruvate (3C), 2 net ATP, 2 NADH

- Does **not** require oxygen (can occur in all cells)
- Glucose is split and oxidized in 10 enzymatic steps

Stage 2: Pyruvate Oxidation (Mitochondrial Matrix)

Input	Output
2 Pyruvate	2 Acetyl-CoA, 2 CO ₂ , 2 NADH

- Bridge step linking glycolysis to the Krebs Cycle

Stage 3: Krebs Cycle (Mitochondrial Matrix)

Input	Output
2 Acetyl-CoA	4 CO ₂ , 6 NADH, 2 FADH ₂ , 2 ATP

- Completes the oxidation of the carbon skeleton from glucose
- All 6 carbons of glucose have been released as CO₂ by this point

Stage 4: Electron Transport Chain & Oxidative Phosphorylation (Cristae)

Input	Output
10 NADH, 2 FADH ₂ , O ₂	~26-28 ATP, H ₂ O

- NADH and FADH₂ donate electrons to protein complexes
- Energy from electrons pumps H⁺ across the inner membrane
- H⁺ flows back through ATP synthase → **chemiosmosis** → generates the most ATP
- **Oxygen** is the final electron acceptor → reduced to H₂O

Total ATP Yield: ~30-32 ATP per glucose (aerobic)

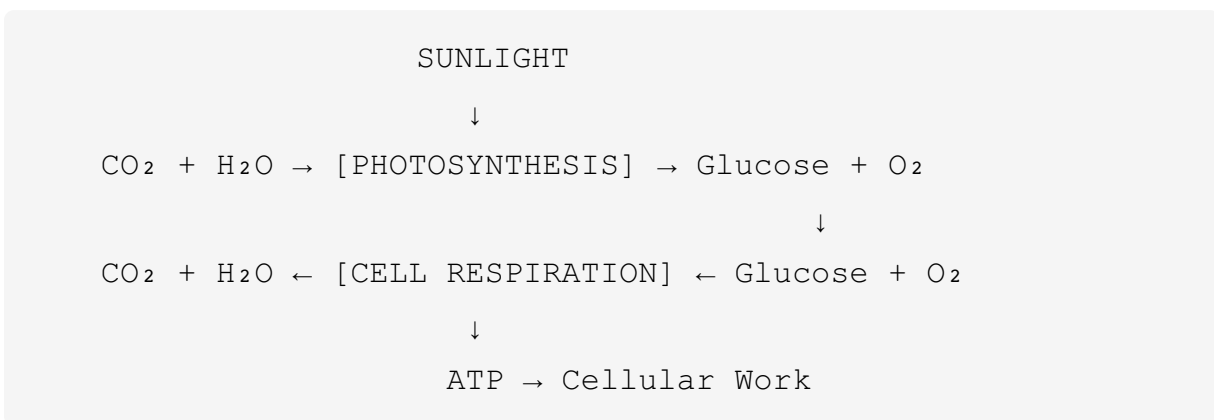
Fermentation (Anaerobic Alternative)

When oxygen is absent:

- Glycolysis still produces **2 ATP** per glucose

- Fermentation regenerates NAD^+ so glycolysis can continue
- Products: **lactate** (animals/bacteria) or **ethanol + CO_2** (yeast)
- Much less efficient than aerobic respiration (2 vs. 30-32 ATP)

Connecting the Two Processes



Both processes use:

- **Electron transport chains** (thylakoid in photosynthesis, cristae in respiration)
- **Chemiosmosis and ATP synthase** (universal mechanism)
- **Redox reactions and electron carriers** (NADPH in photosynthesis, NADH/FADH₂ in respiration)

Study Tips

1. **Master both equations** — They are reverse reactions; if you know one, you know the other
2. **Know the organelles** — Chloroplasts for photosynthesis, mitochondria for respiration
3. **Remember chemiosmosis** — It's the same principle in both: H^+ gradient drives ATP synthase
4. **Trace the electrons** — In photosynthesis: $\text{H}_2\text{O} \rightarrow \text{NADPH}$. In respiration: $\text{NADH/FADH}_2 \rightarrow \text{O}_2 \rightarrow \text{H}_2\text{O}$
5. **Compare the stages** — Make a table: Stage / Location / Inputs / Outputs for all stages
6. **OIL RIG still applies** — Glucose is oxidized in respiration; CO_2 is reduced in photosynthesis

7. **Fermentation is a backup** — Only 2 ATP, but it keeps glycolysis running when O₂ is unavailable